

Intermediate Macro

Chapter 4: Financial Markets I

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Key Tools and Concepts

- Chapter 4 defines **stock** and **flow** variables and distinguishes wealth (a stock) from income (a flow).
- Chapter 4 introduces **monetary policy** and describes **open market operations**.
- Chapter 4 makes use of **balance sheets** for the central bank and private banks.

Assumptions

- Chapter 4 assumes that nominal GDP is given.
- We maintain the previous chapter's assumption that the price level is fixed.
- Money and bonds are the only assets available in the economy.

Determinants of the Demand for Money

4.1 The Demand for Money

- "Semantic Traps: Money, Income, and Wealth"
- Suppose the financial markets include only two assets:
 - ① money, which can be used to purchase goods and services and pays no interest; and
 - ② bonds, which cannot be used for transactions, but pay a positive interest rate i .
- At every moment, households must decide how to **allocate** their given financial wealth **between money and bonds**.
- This decision making problem will depend mainly depend on two variables:
 - ① Your *level of transactions*
 - ② The *interest rate on bonds*

Deriving the Demand for Money

4.1 The Demand for Money

- Let's go from the previous discussion to an equation describing the demand for money.
- Denote the amount of money people and firms want to hold—their *demand for money*—by M^d .
- It seems reasonable to assume that the level of transactions is proportional to nominal income $\$Y$.
- On the other hand, allocating wealth to money comes at the cost of forgone interest rate on bonds. So, money demand should decrease with the interest rate.

Money Demand Function

4.1 The Demand for Money

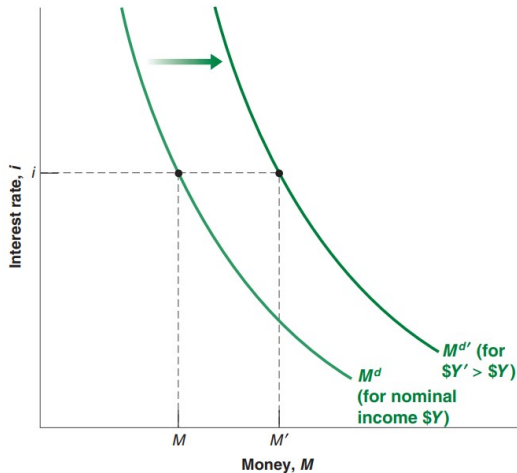
- Putting these observations together, we have

$$M^d = \$Y \times \underset{(-)}{L(i)} \quad (4.1)$$

The demand for money M^d is equal to nominal income $\$Y$ times a decreasing function of the interest rate, with the function denoted by $L(i)$.

M^d Curve

4.1 The Demand for Money



Money Demand, Money Supply, and the Equilibrium Interest Rate

4.2 Determining the Interest Rate I

- Now we look at the supply side of money and then at the equilibrium.
- In this section, assume all money is currency, so there are no checking accounts or private banks.
- Consider the supply of money to be fully in the control of the central bank.

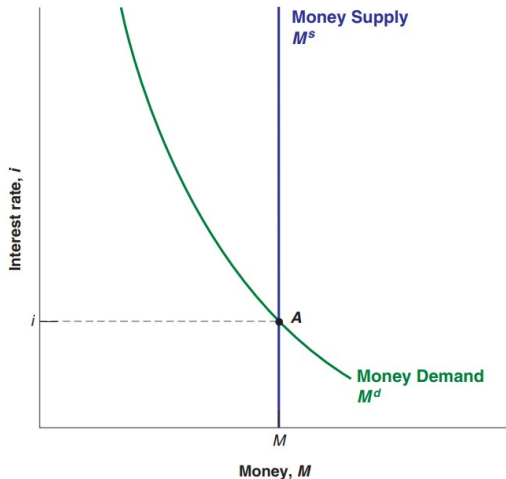
$$M^s = M$$

- The equilibrium condition is

$$\underbrace{M^s = M}_{\text{Money supply}} = \underbrace{M^d = \$Y \times L(i)}_{\text{Money demand}} \quad (4.2)$$

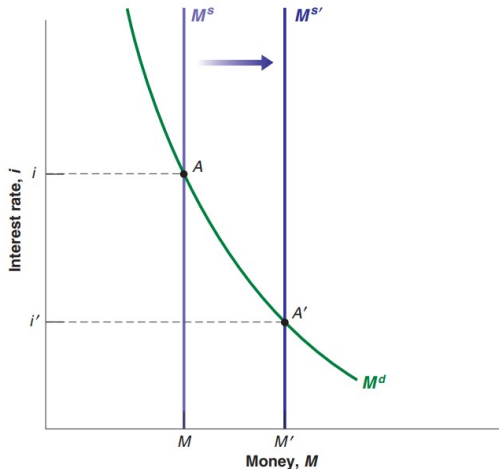
Equilibrium in the Money Market

4.2 Determining the Interest Rate i



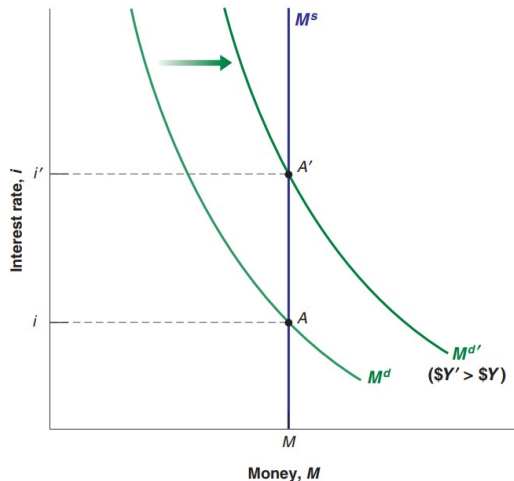
The Effects of an Increase in M^s

4.2 Determining the Interest Rate i



The Effects of an Increase in $\$Y$

4.2 Determining the Interest Rate i



Monetary Policy and Open Market Operations

4.2 Determining the Interest Rate I

- Specifically, how does the central bank control the money supply?
- The way central banks typically change the supply of money is by buying or selling bonds in the bond market. These actions are called **open market operations**.
- Consider the central bank's balance sheet.

Central Bank Balance Sheet	
Assets	Liabilities
Bonds	Money (currency)

The Effects of an Expansionary Open Market Operation	
Assets	Liabilities
Change in bond holdings: +\$1 million	Change in money stock: +\$1 million

Bond Prices and Bond Yields

4.2 Determining the Interest Rate I

- In fact, what is determined in bond markets are not interest rates, but bond *prices*.
- Suppose a bond promises a payment of $\$F$ one year in the future. Call the current price of the bond $\$P_B$. Then, the interest rate (or rate of return) on this bond is given by

$$i = \frac{\$F - \$P_B}{\$P_B}$$

- We can solve this equation for the bond price:

$$\$P_B = \frac{\$F}{1 + i}$$

Given fixed nominal bond payment, *the nominal interest rate and the bond prices are inversely related.*

Federal Reserve Hikes Interest Rate by 0.75%

4.2 Determining the Interest Rate I

- <https://www.youtube.com/watch?v=tDUUiZZt-WA>



Adding Banks into the Model

4.3 Determining the Interest Rate II

- Now we introduce (private) banks into the model. Adding banks to the economy alters the central bank balance sheet only on the liabilities side.

(a)

Central Bank	
Assets	Liabilities
Bonds	Central Bank Money = Reserves + Currency

- Central bank liabilities now consist of currency held by the public plus reserves held by banks. Central bank liabilities—the money the central bank has created—are called **central bank money**.

What Banks Do

4.3 Determining the Interest Rate II

(b)

Banks	
Assets	Liabilities
Reserves Loans Bonds	Checkable deposits

- Banks receive funds from depositors (households and firms) and allow their depositors to write checks against (or withdraw) their account balances. These checkable deposits are liabilities of banks.
- On the asset side, banks hold bonds, loans (which are claims against borrowers), and reserves of some of their deposits.
- Some bank reserves are held in cash and the rest in accounts at the central bank.

The Demand for Central Bank Money

4.3 Determining the Interest Rate II

- Now let us reconsider money market equilibrium in terms of central bank money (H).
- Since bank reserves depend on the amount of checkable deposits we can model this relationship by letting θ represent the **reserve ratio**.
- Recall from our original equation (4.1) that money demand is given by

$$M^d = \$Y \times L(i) \quad (4.3)$$

- Using equation (4.3), the demand for reserves by banks H^d is given by:

$$H^d = \theta \cdot M^d = \theta \cdot \$Y \times L(i) \quad (4.4)$$

Equilibrium in the Market for Central Bank Money

4.3 Determining the Interest Rate II

- The central bank can change the amount of central bank money H through open market operations.
- The equilibrium condition is that the supply of central bank money be equal to the demand for central bank money:

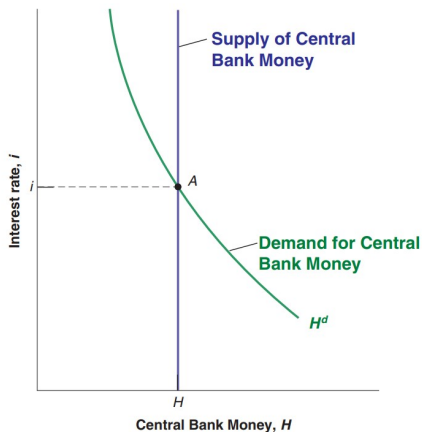
$$H = H^d \quad (4.5)$$

$$H = \theta \cdot \$Y \times L(i) \quad (4.6)$$

Federal Funds Rate

4.3 Determining the Interest Rate II

- The interest rate in the **federal funds market** is called the **federal funds rate**.



Zero Lower Bound Issue

4.4 The Liquidity Trap

- The central bank can choose a desired interest rate by changing the money supply. However, the central bank can push interest rates to zero which would limit further monetary stimulus. This condition is known as a **zero lower bound**.
- When the economy is in this position and monetary policy is no longer effective, the economy is said to be in a **liquidity trap**.
- Increasing the money supply beyond this point has no impact on interest rates.

The Liquidity Trap

4.4 The Liquidity Trap

